

Continuous Assessment 1
Individual Assignment

by

Malcolm Jeremiah Richard

22078778

Programme Name:

BSc (Hons) Information Technology

Semester: May 2026

School of Computing and Artificial Intelligence

Faculty of Engineering and Technology

Sunway University

Video Link: <https://youtu.be/Y5I3UznKLvU>

Contents

1.0	Product Selection and Justification	4
1.1	Interface Identification and the Target Audience	4
1.2	Primary User Tasks	4
1.3	Motivation and Usability Concerns	5
2.0	Usability Framework Selection and Justification	6
3.0	Heuristic Evaluation Log	7
	Violation V01: Match Between System and Real World (H2) – Severity: 2	7
	Figure 1: Profile page displaying enrolment jargon alongside user’s name	7
	Violation V02: System Mapping Leak (H5) – Severity: 2	8
	Figure 2: Contact Information page showing enrolment jargon as ‘Last Name’	8
	Violation V03: Critical Lifestyle Filtering Failure (H4) – Severity: 3	9
	Figure 3: ‘Completed Courses’ activated but not showing anything	9
	Violation V04: Active Sorting Directory Clutter (H7) – Severity: 3	10
	Figure 4 & 5: ‘Courses I am taking’ and ‘Current Courses’ activated but none is working	10
	Violation V05: Aesthetic and Minimalist Design (H8) – Severity: 2	11
	Figure 6: Showcasing the same artwork card used among majority of the subjects	11
	Violation V06: Recognition Rather than Recall (H6) – Severity: 1	12
	Figure 7: Global left menu panel relocating to top right upon entering messages page	12
	Violation V07: Architectural Layout Customization (H3) – Severity: 2	13
	Figure 8: Empty Organizations page	13
	Violation V08: Visibility of System (H1) – Severity: 2	13
	Figure 9: Messages tab showcasing 99+ messages with zero knowledge of urgency	13
	Violation V09: Help and Documentation (H10) – Severity: 2	14
	Figure 10: Course module card consisting of enrolment/technical jargon	14
	Violation V10: Error Prevention for Users (H10) – Severity: 2	15
	Figure 11: Submission upload page	15
4.0	User Validation	16
4.1	Data Triangulation	16
4.2	Analysis of the Gaps	17
5.0	Redesign Recommendations	18
5.1	Redesign 1	18
	Figure 12: Proposed profile landing page with an edit option and technical/enrolment information shown in detail neatly	18
5.2	Redesign 2	19

Figure 13: Proposed contact information card, where editing first, last and additional name is allowed under admin moderation.....	19
5.3 Redesign 3.....	20
Figure 14: Proposed course directory page, where the search bar and filters are functional	20
5.4 Redesign 4.....	21
Figure 15 & 16: Proposed messages button and hub to increase urgency and ease of use	21
5.5 Redesign 5.....	22
Figure 17: Proposed 'Info' button to help new students on understanding their course	22
6.0 Reflection	23
7.0 AI Use Statement	23
8.0 Reference	24

1.0 Product Selection and Justification

1.1 Interface Identification and the Target Audience

The digital interface which I have selected for this structured heuristic evaluation would be Sunway University's institutional learning management platform, which is the eLearn portal which is powered by Blackboard Ultra. This site uses the URL of *elearn.sunway.edu.my*. This production, level enterprise application come website serves as the primary hub for academic workflow orchestration, lecture material sharing, student-faculty communication and continuous grading assessments. The primary target user's demographic consists of undergraduate students, specifically within the School of Computing and Artificial Intelligence and the Faculty of Engineering and Technology. This user base was selected as we would share a similar experience considering we're in the same school and faculty, students who rely on the platform many times weekly to manage their active courses. Concurrently, this interface supports academic staff, lecturers and admin staff who uses the system for lecture content sharing purposes, cohort management and grade book publications. The eLearn system operates as a deeply integrated single-page architecture which features more than five different navigations routes on its sidebar, which includes the global Activity Stream, User Profile, Courses Directory, Organizations Panel, Calendar, Messages Hub, Grades Dashboard and System Tools workspace.

1.2 Primary User Tasks

To evaluate the eLearn's interface with empirical clarity, three user workflows focusing on regular administrative interactions have been planned out. These tasks are intentionally planned out to serve as a diagnostic operational track, which forces users to navigate through the website pathways and directly test the interface's structural logic against design flaws visible within the eLearn system.

Task 1: The location and extraction of active institutional announcements.

Driven by routine, students need to monitor campus updates. Thus, this task requires the user to access the Activity Stream page and scroll down the central chronological timeline to locate specific institutional announcements or notices. This task evaluates how easily a student can pick out important and urgent academic or institutional updates when they are mixed into a

massive stream of general campus noise, directly testing eLearn's content hierarchy and layout design.

Task 2: Navigating Sorting Filters inside the Courses directory

This task workflow involves a student, navigating to the Courses tab and utilizing the built-in filter option which is located at the top of the page, to switch the view from Courses from various years, courses completed, or ongoing courses. This task does not require accessing any locked course content. Instead, this task is purely being carried out to evaluate the internal sorting consistency and logic of the platform's directory categorization.

Task 3: Tracking module Messages channel under high notification volume

This task is triggered by a prominent notification anchor showing an overwhelming update which says, "Messages 99+". The user must enter the centralized Messages hub to locate a specific message from a specific instructor. The user must then scan through various course rows, identify specific active modules and click into the folder layout. This task checks layout uniformity within the messaging hub when a student is under a high notification volume.

1.3 Motivation and Usability Concerns

The choice of Sunway University's eLearn portal, which is powered by Blackboard Ultra, is driven by a production-level operational friction which was observed during daily student-related tasks rather than just visual complaints. While this website/application successfully utilizes contemporary minimalist design rules to present a modern visual, a deeper heuristic inspection revealed multiple friction and pain points where certain layouts and design actually compromise task efficiency.

A primary example stating the case would be the global 'Messages 99+' badge found on the left side of the webpage, which could potentially produce significant notification anxiety for students. While students can click into the inbox folder to view their inbound communication matrix, the primary navigation layers fail to provide an upfront filtering system. The system

stacks every unread notice and updates into a massive, uniform global count whether urgent or not, forcing users to execute repetitive guesswork clicks across multiple layout menus to gauge message priority.

Furthermore, exploring these workflows exposes distinct layout constraints when students attempt to monitor everyday campus updates. While the platform effectively structures its chronological timeline to ensure important notices are pinned or fly to the top of the Activity Stream feed, it frequently fails to maintain clean information design standards. High-priority bulletins are regularly diluted with unrendered image file path markers and lengthy, unformatted walls of raw hyperlink data strings, causing immediate visual scanning fatigue.

Similarly, the platform's categorization rules present severe internal contradictions; the course filtering drop-down menu completely fails to list archived, completed modules under the designated "Completed Courses" tab, rendering the sorting widget non-functional and forcing students into lengthy manual scrolling routines past locked course cards. Finally, navigating into individual communication threads triggers a jarring layout regression to not styled legacy table views where interactive command buttons remain completely non-responsive and fail to provide clear system feedback when pressed.

2.0 Usability Framework Selection and Justification

This report deploys the Nielsen's 10 Usability (1994) as its core diagnostic model. Nielsen's guidelines give us an ideal analytical model in aspects such as stateful, complex enterprise software because they are more focused on mapping clarity, visual consistency, user freedom and structural prediction. Another alternative design framework like the Schneiderman's Eight Golden Rules, focuses more on macro-level software behaviours and gives a lesser precise guidance on single-page variations. Nielsen's framework allows for a more accurate analysis of specific single-page errors such as unresponsive control elements, internal visual regressions and wrongfully adjusted category filters. To preserve methodology clarity during the data triangulation phase, no external design metrics will be introduced in this study. Every identified flaw and layout defect will be systematically cross-referenced against Nielsens definitions and scored on a strict scale of 0 to 4 severity, where 0 being a non-issue and 4 being very severe, which establishes an absolute, logical link between identified flaws and subsequent low-fidelity interface redesigns.

3.0 Heuristic Evaluation Log

A thorough audit of the eLearn portal was carried out using the user's active system screenshots as empirical evidence. In the study below, 10 comprehensive architectural and structural violations were identified across the e-Learn system.

Violation V01: Match Between System and Real World (H2) – Severity: 2

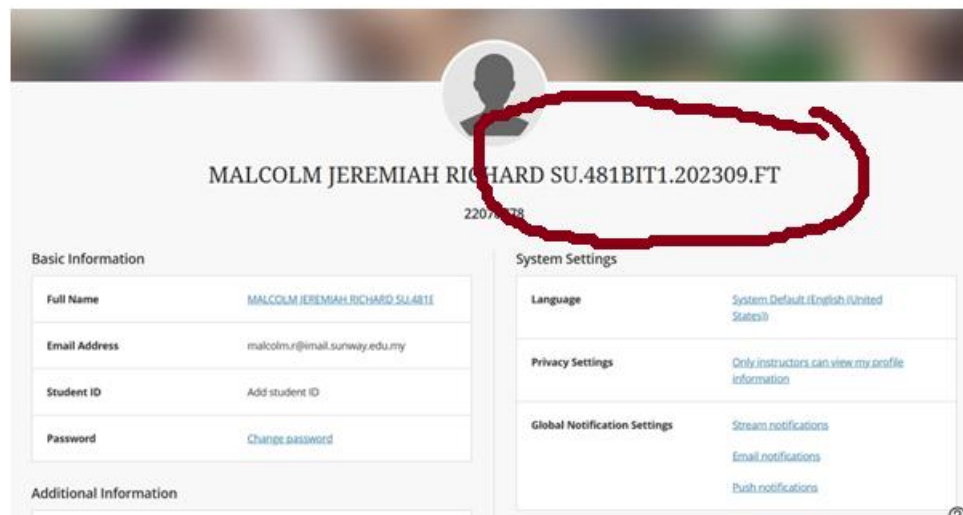


Figure 1: Profile page displaying enrolment jargon alongside user's name

This flaw represents a severe breakdown in real-world mental mapping. In professional enterprise designs, system-facing database keys and backend enrolment strings should be completely masked behind an interface presentation layer. However, in the eLearn platform, it was leaked directly the user's profile header '*MALCOLM JEREMIAH RICHARD SU.481BIT1.202309.FT*' in identical font size, like weight and the same font colour in which the system treats an administrative database tracking key. This structural choice forces the user to scan past developer syntax to verify their identity.

Violation V02: System Mapping Leak (H5) – Severity: 2

1.2

Profile Settings

Contact Information

Basic Information

First Name
MALCOLM JEREMIAH RICHARD

Last Name
SU.481BIT1.202309.FT

Additional Name
MALCOLM JEREMIAH RICHARD

Email Address
malcolm.r@imail.sunway.edu.my

Figure 2: Contact Information page showing enrolment jargon as 'Last Name'

This flaw was discovered upon discovering the “Contact Information” settings side panel which exposes a lack of defensive data entry boundaries. The platform’s registration backend had hard-coded the user’s intake information ‘*SU.481BIT1.202309.FT*’ in to the legal ‘Last Name’ field. Because of this, the global interface template automatically combines the ‘First name’ and ‘Last Name’ fields to output titles across class rosters, logs and everything else which causes the student to be branded with the technical system intake data. Because this input text box is not editable by students, the user is trapped in that system state with zero freedom to resolve it. It makes absolute sense in the eyes of a database administrator, however from a UI/UX engineering perspective, it is a massive violation.

Violation V03: Critical Lifestyle Filtering Failure (H4) – Severity: 3

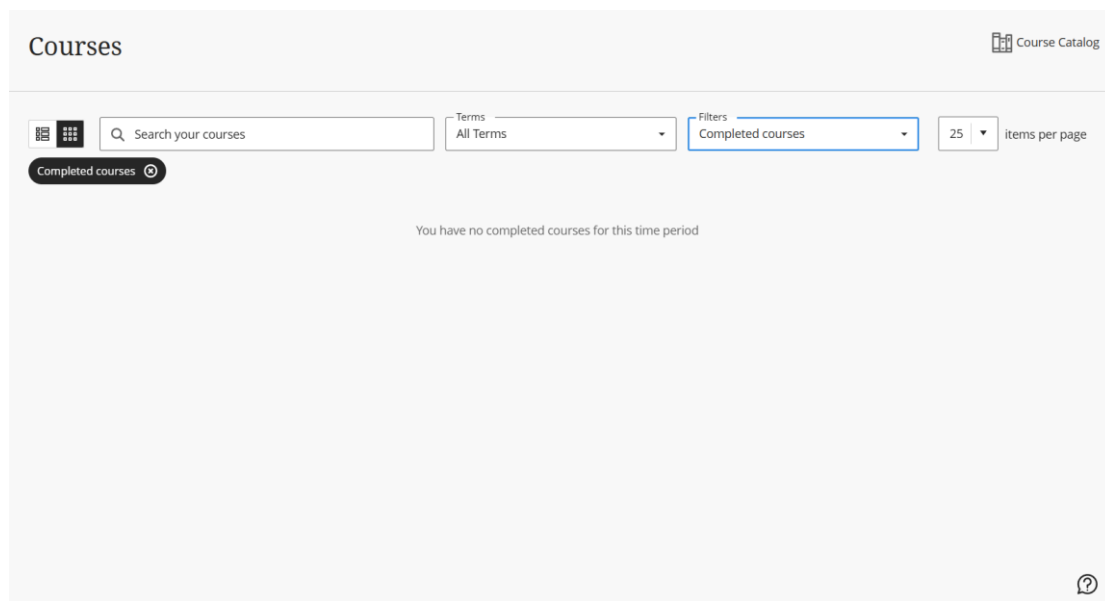


Figure 3: ‘Completed Courses’ activated but not showing anything

This structural design flaw showcases a major contradiction within the platform’s course directory sorting and filtration system. When a student adjusts their active filter configuration to ‘Completed Courses’, they expect to have a proper view of all the courses they had completed in previous semesters in a historical, finalized and neat style. Instead, Blackboard’s sorting algorithm returns a completely blank page stating ‘You have no completed courses for this time period’ which directly contradicts the reality of the dashboard which is heavily cluttered with expired and locked classes from the past semesters. Breaking this logical grouping hinders consistency which renders the dropdown filter useless.

Violation V04: Active Sorting Directory Clutter (H7) – Severity: 3

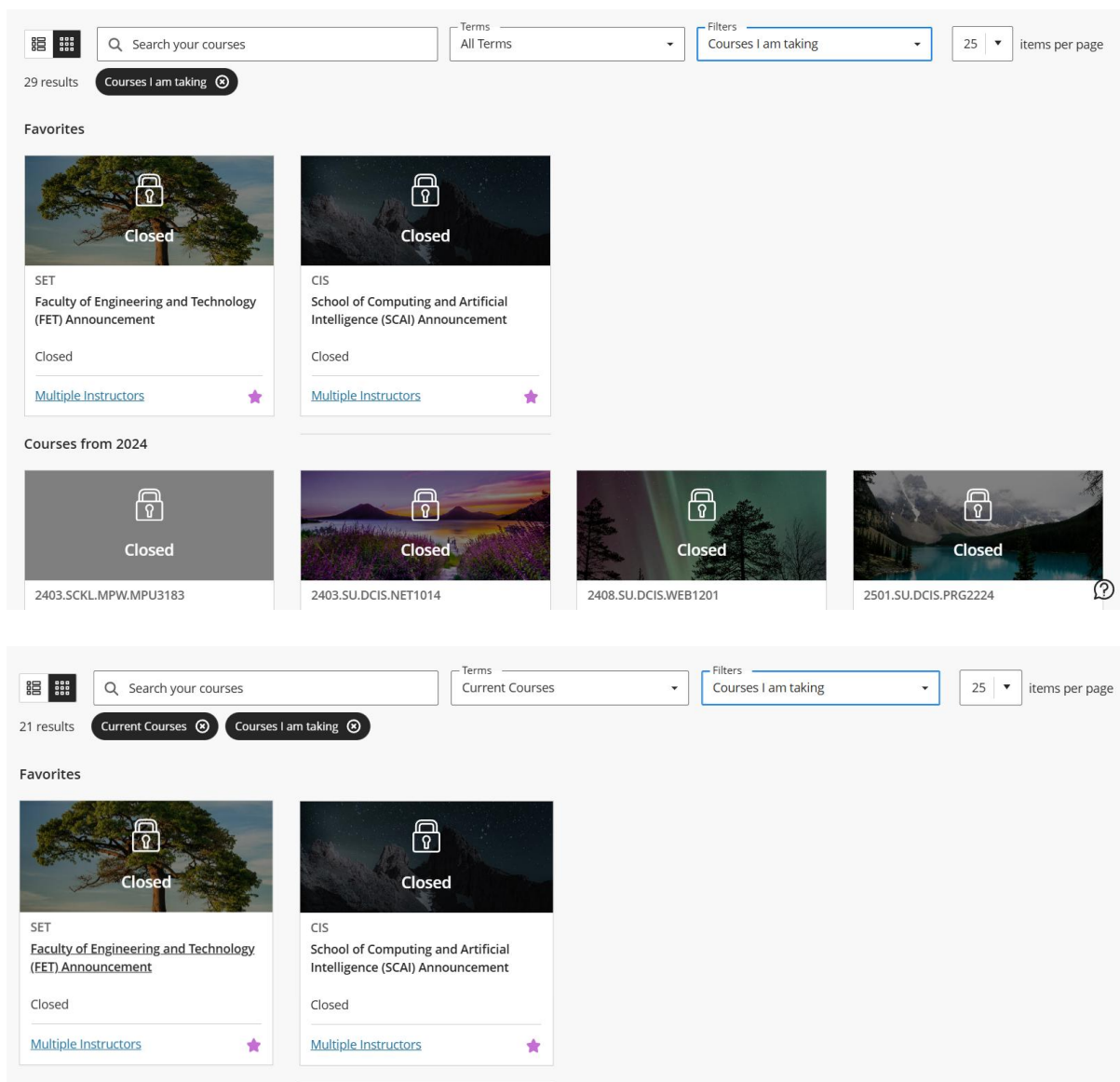


Figure 4 & 5: 'Courses I am taking' and 'Current Courses' activated but none is working

Violation 4 surfaces as a severe efficiency bottleneck in the primary course list directory. This happens when a student activates the 'Courses I'm taking' filter in the dropdown menu to isolate the active modules which requires attention and interaction. However, the system's sorting logic breaks the logic by flooding the top view with locked courses from years before. Because blackboard fails to provide users with the information needed based on the filter dropdown option and does not offer a way to archive or delete previous course blocks in the course directory, the workspace gets crowded and students have to scroll past many dead units every day to get to their desired destination which slows down task efficiency.

Violation V05: Aesthetic and Minimalist Design (H8) – Severity: 2

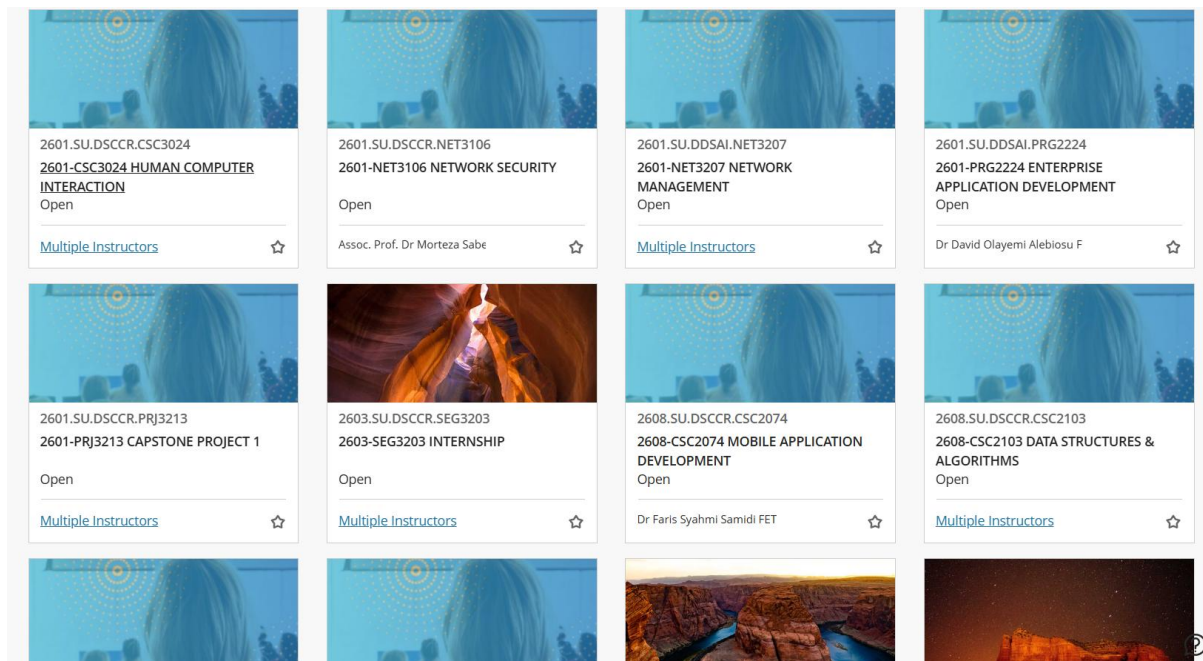


Figure 6: Showcasing the same artwork card used among majority of the subjects

This visual design failure happens across the course directory in which the platform applies the same exact abstract art card asset across completely different academic course modules. This total lack of visual variety and options deprives the user interface of different art works which induces visual scanning fatigue for the student. Because human eyes rely a lot on unique colour coding boundaries to rapidly scan interactive screen nodes, the repeating art card on most of the course modules forces the students to rely on slowly reading the titles on each card one-by-one just to locate a specific class, which violates minimalist design principles.

Violation V06: Recognition Rather than Recall (H6) – Severity: 1

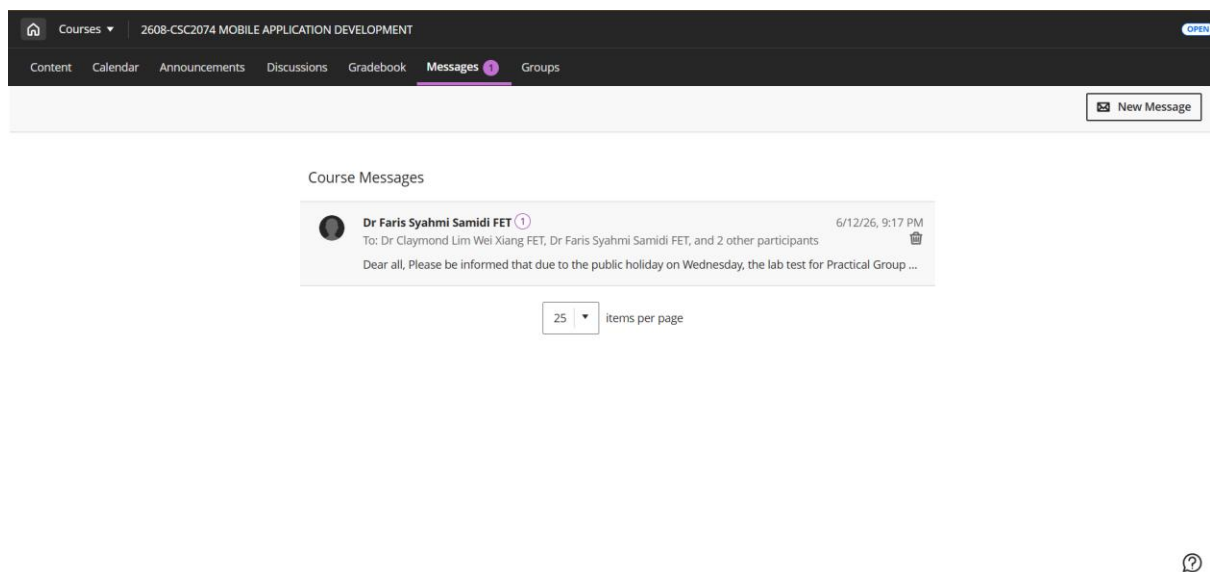


Figure 7: Global left menu panel relocating to top right upon entering messages page

This structural defect showcases a breach of internal application navigation in which the single-page web application which holds the primary global navigation menu (Activity Stream, Courses, etc) should remain persistent and anchored to the original location of the screen which is the left-hand side of the screen. However, the moment a student clicks into a specific course block, the system completely removes the left side navigation bar and is relocated to the top. Hijacking the primary navigation bar breaks basic design consistency standards and traps users inside deep sub-menu areas which causes spatial disorientation for some.

Violation V07: Architectural Layout Customization (H3) – Severity: 2

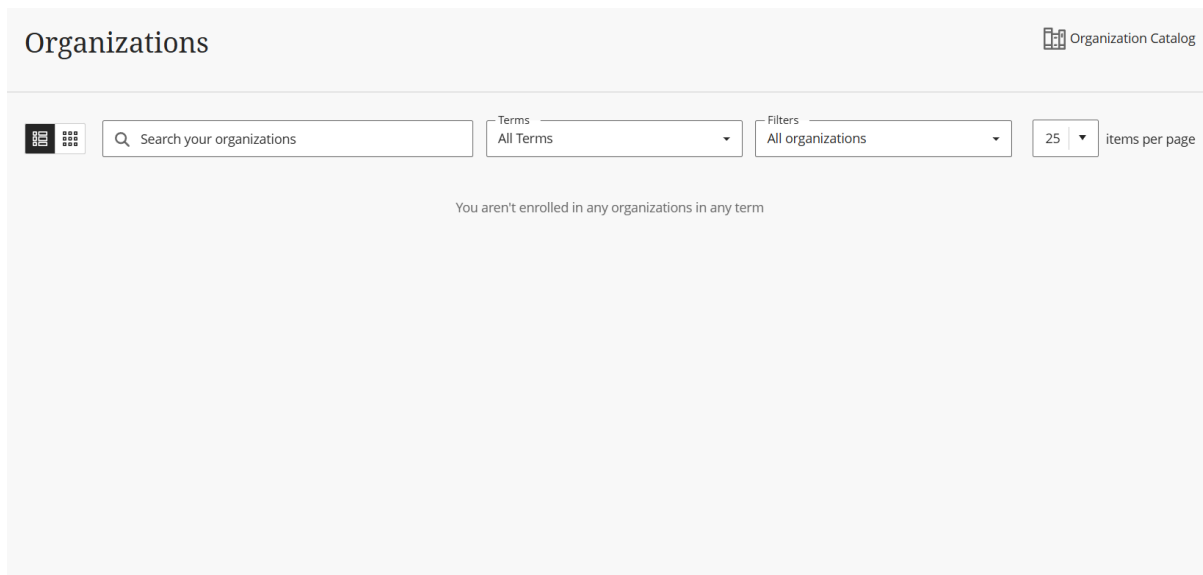


Figure 8: Empty Organizations page

This interface issue manifests inside the centralized workspace of the platform, specifically the Organization's page. For a student who is not involved in any organizational affiliations on e-Learn, the panel loads a completely blank white page. However, the main navigation layout provides no way for a student to configure or customise in which students can interact with hide toggles or customization tools, which forces a permanent, nonfunctional button onto a high-traffic system which restricts user control and freedom, potentially cluttering the main navigation menu with unused elements.

Violation V08: Visibility of System (H1) – Severity: 2

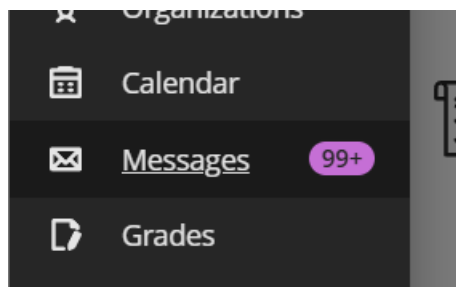


Figure 9: Messages tab showcasing 99+ messages with zero knowledge of urgency

The persistent Messages tab icon presents a flat notification alert badge showing 99+. Essentially, it is blending all critical private module instructions from active lecturers with

student club broadcast or in other words, less important matters together into a single massive bucket counter, the system fails to display true system status. The user cannot judge the urgency of the messages in that pool or know of an important notice without clicking through multiple sub-directories.

Violation V09: Help and Documentation (H10) – Severity: 2

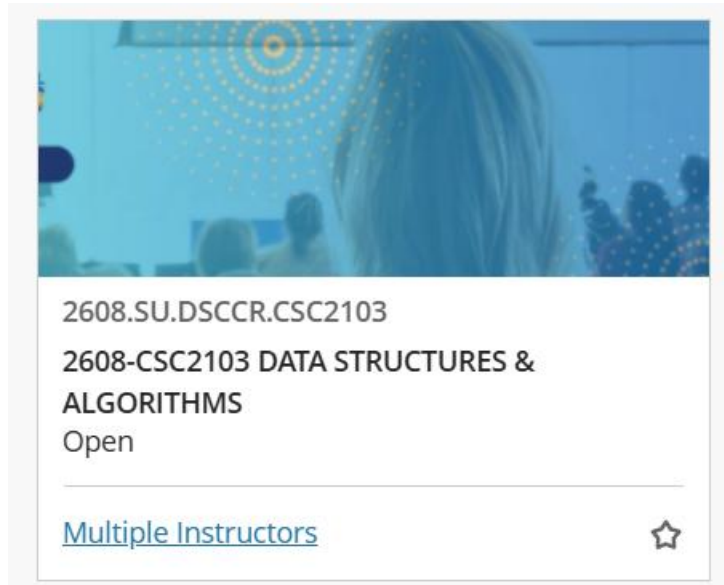
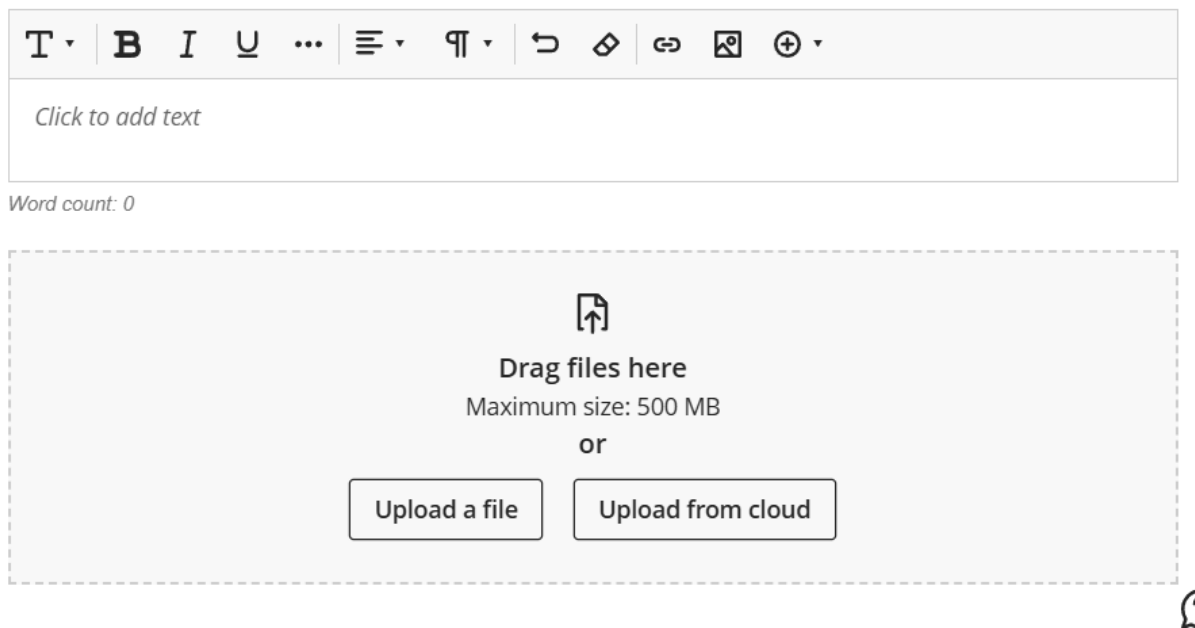


Figure 10: Course module card consisting of enrolment/technical jargon

Every course module card component in e-Learn forces a dense, backend data string entry array such as 2608.SU.DSCCR.CSC2074 right onto the card. Although these strings are related to our course information, there are students who sometimes find it difficult to decipher what it really means. The interface fails to provide any quick hover-tips or an info (“i”) on the card explaining what certain elements mean. Forcing raw technical shortcuts onto high-traffic screens without any help leaves some users clueless.

Violation V10: Error Prevention for Users (H10) – Severity: 2

Submission



The screenshot shows a submission interface. At the top is a toolbar with icons for text formatting (bold, italic, underline, bulleted list, numbered list, link, unlink, image, and a plus sign). Below the toolbar is a text input area with the placeholder text "Click to add text". Underneath the text area, it says "Word count: 0". The main part of the interface is a large dashed box containing a file upload icon (a document with an upward arrow), the text "Drag files here", "Maximum size: 500 MB", and the word "or". Below this are two buttons: "Upload a file" and "Upload from cloud".

Figure 11: Submission upload page

Every course submission container in e-Learn isolates itself from the actual assignment requirements, which causes a total lack of error prevention guidance. As of now, the upload bubble does not analyse the assignment parameters to remind students which file type is needed for the assignment. Often times, some assignments offer only one submission which is highly terrifying for students especially if they make a mistake (e.g. submitting a PDF file instead of a DOCX file). The platform misses a massive opportunity to display a simple reminder for the students, whether it being manually inserted by the lecturer or an automated system, which file type is required for the assignment. A little reminder such as “Remember, this assignment requires a ZIP file!” will help forgetful students. Leaving a submission page generic, forces students to rely on manually checking the file, which can sometimes cause accidental submission errors, which causes them to have back and forth conversations with their lecturers to reverse the mistake.

4.0 User Validation

To empirically support and analyse the flaws encountered, a think-aloud study was executed with 3 independent students, running the diagnostic operations detailed in 1.2.

4.1 Data Triangulation

ID	Heuristic	Student Status (Confirmed)	Real-World Behaviour	Severity (0-4)
V01	H2: System and World	3/3	Users mentioned that adding the enrolment shortcuts into the name causes the account profile page to seem cluttered	2
V02	H5: Error prevention	3/3	Users noticed that the enrolment keys were hard coded into the uneditable last name field, calling it weird and unpolished	2
V03	H4: Consistency	2/3	Users clicked “Completed Courses” and was met with an empty layout, confused as they were in their final year already	3
V04	H7: Efficiency	3/3	Users expressed friction scrolling past locked and irrelevant course cards to get to their destination	3
V05	H8: Aesthetic Design Purposes	2/3	Users showcased exhibited visual tracking delay and admitted that they rely on the titles, rather than the art cards	2
V06	H6: Recognition vs Recall	1/3	Users noted that hiding the primary left sidebar was	1

			annoying, but was not much of an issue	
V07	H3: User control	2/3	Users noted that keeping a blank page locked onto their navigation menu with no ability to customise is frustrating	2
V08	H1: System Status	3/3	Users state that the flat “99+” gave them zero clue on whether there’s an urgent message needing attention	2
V09	H10: Help and Documentation	3/3	Users admit the lack of any tooltips or guides explaining certain aspects of the platform	2
V10	H9: Error Recovery	2/3	Users suggested a small popup reminder for students to ease submission	2

4.2 Analysis of the Gaps

The data triangulation established a clear relation between the system-centric database mapping decisions and active user friction. While prior evaluation categorized some violations as minor formatting flaws, empirical behavioural observation confirmed that some are noticeable structural defects, in which supported by the students. Crucially, validation confirmed that for example, the 99+ badge causes anxiety as users are not aware of the urgency of the messages.

5.0 Redesign Recommendations

5.1 Redesign 1

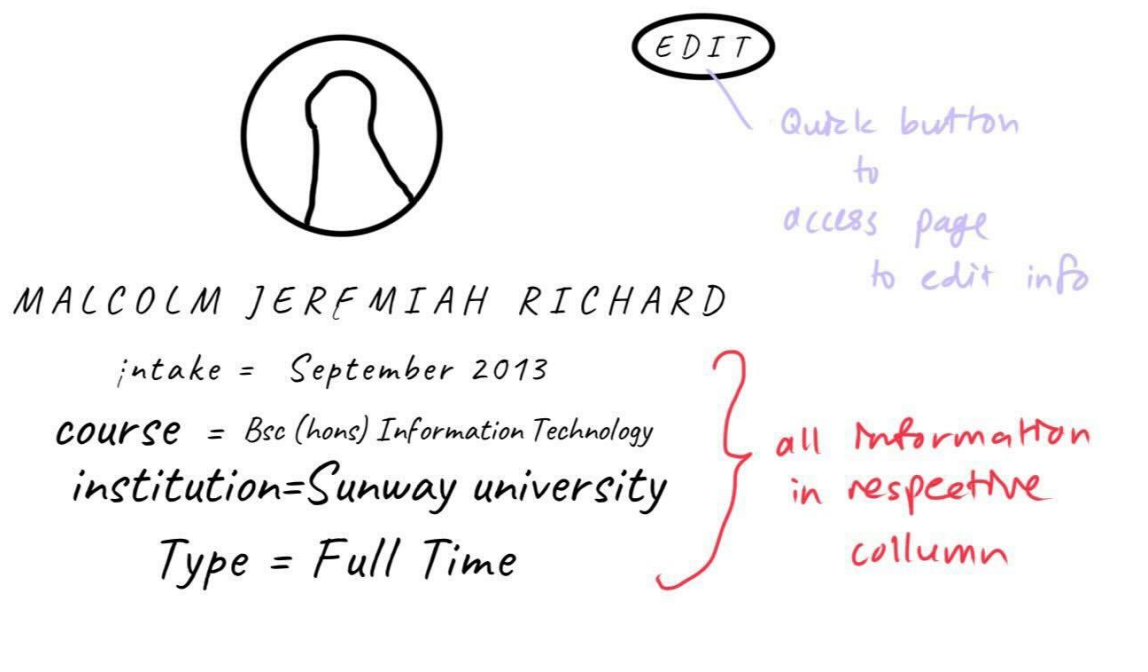


Figure 12: Proposed profile landing page with an edit option and technical/enrolment information shown in detail neatly

UX rationale: Moving the raw enrolment details out of a prominent display text fields helps restore natural mapping matching with real-world human identity standards over system database strings (Nielsen, 1994). Grouping all the enrolment data such as course and intake into structured data rows matches a student's mental model while having an edit option for users to edit, which stops entry leaks without breaking database tracking logs.

Trade-offs: This method requires development effort to modify the front-end of the webpage and engineer a request pipeline routing edits to the admin. It also demands a slightly more vertical screen space.

5.2 Redesign 2

☐ Profile settings

Contact Information

Basic Information

First Name
MALCOLM JEREMIAH RICHTARD

Last Name
—

Additional Name
—

edit button giving users control over their information (reviewed by admin)

Figure 13: Proposed contact information card, where editing first, last and additional name is allowed under admin moderation

UX Rationale: To support the display changes introduced in Redesign 1, the layout proposed applies the core principles of user control and freedom (H3) by giving students the ability to edit text form components such as the first, last and additional name details. (Rogers et al., 2011). Separating inputs in this box and attaching the pencil symbols lets users fix database entry errors while maintaining an admin moderation checking to preserve appropriate names.

Trade-offs: Increase development scope to build the front-end pipeline and code input boxes as stateful, responsive interactive fields instead of flat and static texts.

5.3 Redesign 3

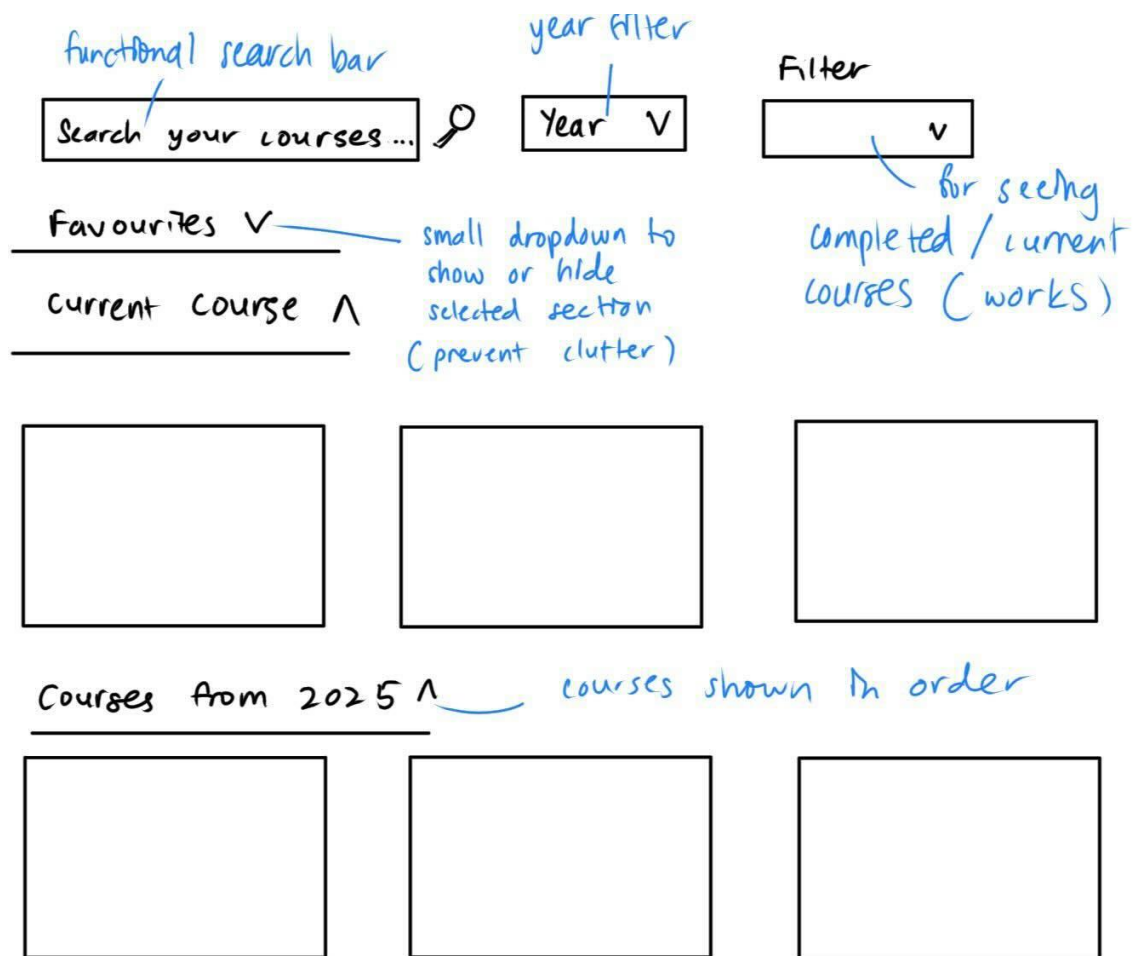


Figure 14: Proposed course directory page, where the search bar and filters are functional

UX Rationale: This dashboard addresses efficiency of use (H7) by including the robust search bar, year selector and dropdown filters that work onto the menu bar. Other than that, segmenting course modules into collapsible blocks helps reduce the clutter in the course directory page, helping users find their destinations with ease.

Trade-offs: Increase in technical development complexity to cache each user preferences locally within browser storage. The header area also consumes minor vertical screen space.

5.4 Redesign 4

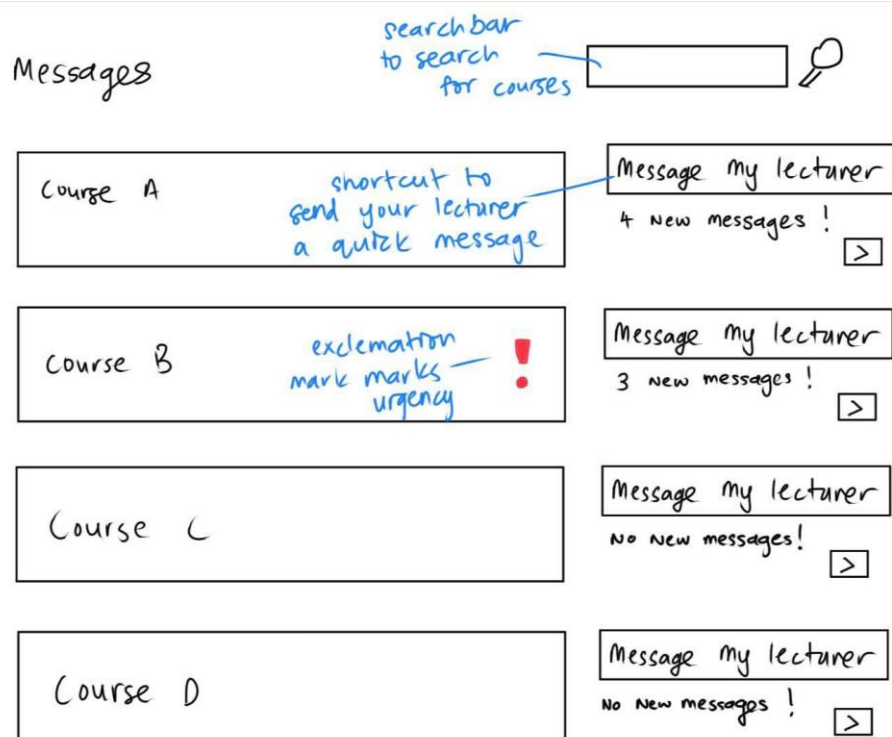
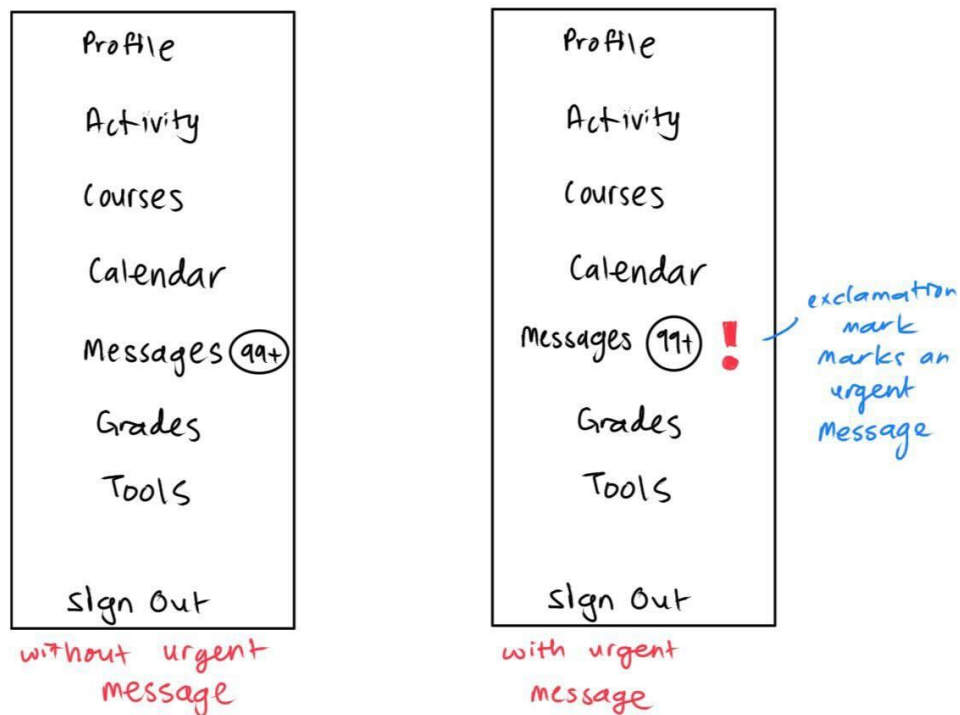


Figure 15 & 16: Proposed messages button and hub to increase urgency and ease of use

UX Rationale: This messaging layout improves the visibility of the system status (H1) by changing up the flat notification metrics into clean, priority-based visual alert cues (Shneiderman & Plaisant, 2010). Placing an urgent exclamation mark badge next to certain courses signal immediate priorities which helps student know where to pay their attention too rather than aimlessly scrolling through all the messages. Inside the hub, there is also a Message my lecturer shortcut button to make it easier for students to communicate with their teachers which can also reduce memory recall loads during high notification volume periods.

Trade-offs: This method demands higher backend logic to rank the incoming message data into urgency tiers.

5.5 Redesign 5

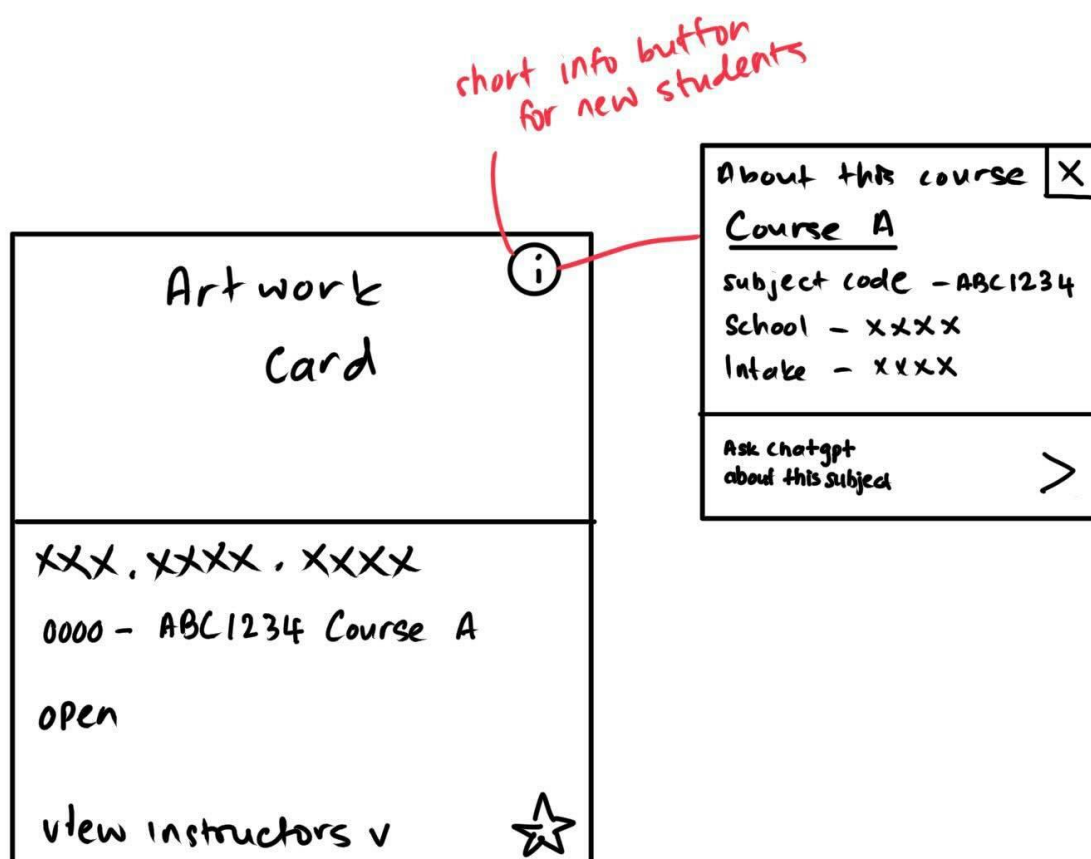


Figure 17: Proposed 'Info' button to help new students on understanding their course

UX Rationale: This card modification satisfies the help and documentation standards (H10) by providing a short and small info box. A plain-language text detail box helps students who

experience complex jargon (Benyon, 2014). The interactive info button keeps the dashboard minimalist while letting new students review the course details as well as potentially ask an AI for help.

Trade-offs: Client-side scripting will be more complex to handle modal overlay triggers from course API. Integrating a chatbot also expands the deployment scope.

6.0 Reflection

After evaluating an interface where I rely on a lot highlighted how heavy user familiarity can decently mask design flaws within the system. Although the e-Learn system is good enough for most students, my own muscle memory had built workarounds for blatant layout flaws such as the disappearing left bar (V06) and the forced scrolling to find my course modules. Relying on Nielsens heuristic was important in this study to reset this bias, evaluating the portal objectively rather than relying on normal habits.

However, the study's sample size being only 3, limits its generalizability across the wider university cohort. More technical students can adapt to system-centric jargon much quicker and efficiently compared to non-technical students. To expand this study and research effectively, I would incorporate a large-scale quantitative clickstream checking to be actually able to trace actual manoeuvre rates across the entire system, replacing subjective verbal feedback, with concrete statistical data.

7.0 AI Use Statement

In the process of completing this study, an artificial intelligence language model (Gemini) was utilized as an assistive tool. The tool was exclusively used to optimize grammatical flow, refine professional tone and format the entire report properly. All core product selections, primary user tasks, heuristic violation tracking and evidence-based redesign concepts remain the original work of me.

8.0 Reference

Benyon, D. (2014). *Designing interactive systems: A comprehensive guide to HCI, UX and interaction design* (3rd ed.). Pearson.

Nielsen, J. (1994). *Usability engineering*. Morgan Kaufmann.

Rogers, Y., Sharp, H., & Preece, J. (2011). *Interaction design: beyond human-computer interaction* (3rd ed.). John Wiley & Sons.

Shneiderman, B. (2010). *Designing the user interface: Strategies for effective human-computer interaction* (5th ed.). Pearson Education India.